

WORKSHEET - NAPLAN Year 9

Number and Algebra Ratios, Scale and Proportion (9)

Teacher: Joanne Rust

Exam Equivalent Time: 10 minutes (based on NAPLAN allocation of ~1.25 min/mark)



Questions

1. ☒ Number and Algebra, NAP-66187

The picture below shows a flower.



The picture is 2 cm wide. The actual flower is 60 cm wide.

What scale is used in the picture?

- ☐ 2 cm represents 20 cm
☐ 4 cm represents 30 cm
☐ 1 cm represents 30 cm
☐ 1 cm represents 3 cm

2. ☒ Number and Algebra, NAP-60268

In a science experiment, Albert needs to add 60 millilitres of acid to every 2 litres of water.

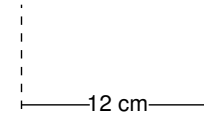
If Albert only has 0.5 litres of water left, how many millilitres of acid should he add?

- 0.15 mL** **0.6 mL** **15 mL** **30 mL** **120 mL**
☐ ☐ ☐ ☐ ☐

3. ☒ Number and Algebra, NAP-42667

The actual body length of a beetle Brad has caught is 24 mm.

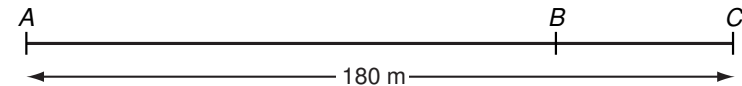
A scale drawing of the beetle is shown below.



What scale is used in the drawing?

- ☐ 1 cm represents 5 mm
☐ 1 cm represents 2 mm
☐ 2 cm represents 1 mm
☐ 5 cm represents 1 mm

4. ☒ Number and Algebra, NAP-01597



The distance from **A** to **B** is three times the distance from **B** to **C**.

The distance from **A** to **C** is 180 metres.

What is the distance from **B** to **C**?

- 30 metres 40 metres 45 metres 60 metres
☐ ☐ ☐ ☐

5. ☒ Numbers and Algebra, NAP-36810

In a tap dancing class, the ratio of boys to girls is 5:7.

There are 6 more girls than boys in this group.

How many students are in this group altogether?

- 12** **13** **18** **36**
☐ ☐ ☐ ☐

6. ✕ Number and Algebra, NAP-30867

Paul stacks milk cartons into supermarket refrigerator shelves.

Each shelf is stacked with 6 full cream milk cartons, 4 lite milk cartons and 2 skim milk cartons.

Every hour Paul stacks 240 milk cartons in total.

How many lite milk cartons does he stack every hour?

7. ✕ Number and Algebra, NAP-72189

Marshall keeps a tin of lollipops on his desk at home.

His son Dallas takes 1 lollipop out every 3 days.

Marshall adds a new lollipop to the tin every 4 days.

The lollipops in Marshall's tin are decreasing by 1 lollipop every x days.

What is the value of x ?

8. ✕ Number and Algebra, NAP-42752

Suzie rents a caravan from a hire company.

The cost to rent the caravan is directly proportional to the number of kilometers driven.

The rental cost to drive 200 km is \$112.50.

What is the rental cost to drive 60 kilometres?

\$

Worked Solutions

1. Number and Algebra, NAP-66187

1 cm represents 30 cm

2. Number and Algebra, NAP-60268

Given 60 mL $\Rightarrow$ 2 litres

30 mL $\Rightarrow$ 1 litres

15 mL $\Rightarrow$ 0.5 litres

3. Number and Algebra, NAP-42667

Scale:

12 cm : 24 mm

120 mm : 24 mm

5 : 1

In the options given, the ratio of 5 : 1 occurs when

1 cm represents 2 mm

4. Number and Algebra, NAP-01597

$AB : BC$

3 : 1

Since $AC = 180$ metres,

$AB = 135, BC = 45$

5. Numbers and Algebra, NAP-36810

5:7 means that given 12 students,

$\Rightarrow$ 5 boys, 7 girls

If 24 students,

$\Rightarrow$ 10 boys, 14 girls

If 36 students,

$\Rightarrow$ 15 boys, 21 girls

$\therefore$ In a class of 36 students, there are 6 extra girls.

6. Number and Algebra, NAP-30867

Shelves stacked each hour

$$= \frac{240}{12} = 20$$

$\therefore$ Lite milk cartons stacked

$$= 20 \times 4$$

$$= 80 \text{ per hour.}$$

7. Number and Algebra, NAP-72189

**Every 12 days, Dallas takes 4 lollipops
and Marshall adds 3.**

$$\therefore x = 12$$

8. Number and Algebra, NAP-42752

Rental cost per km

$$= \frac{112.50}{200}$$

$$= 56.25 \text{ cents/km}$$

$\therefore$ Cost to drive 60 km

$$= 60 \times 56.25$$

$$= 3375 \text{ cents}$$

$$= \$33.75$$